

CE 211 – Concrete Technology Cheat Sheet

Ingredients of Concrete

- Cement (binder): OPC, PPC, PSC, blended cements.
- Water: must be potable; w/c ratio controls strength & durability.
- Aggregates: coarse (strength, bulk), fine (workability, filling).
- Admixtures: chemical (plasticizers, retarders, accelerators, air-entrainers); mineral/SCMs (fly ash, GGBS, silica fume, RHA).

Cement Chemistry

- Main compounds: C3S (early strength), C2S (late strength), C3A (fast set, low sulfate resistance), C4AF (minor role).
- Gypsum (~5%): prevents flash set by controlling C3A.
- Hydration: $C_3S, C_2S + H \rightarrow C-S-H$ (strength gel) + $Ca(OH)_2$.
- Pozzolanic reaction: $SCM + Ca(OH)_2 + H \rightarrow$ extra $C-S-H$.

Bogue's Equations

- Estimate compound % from oxide composition.
- $C_3S \approx 4.07CaO - 7.60SiO_2 - 6.72Al_2O_3 - 1.43Fe_2O_3 - 2.85SO_3$.
- $C_2S \approx 2.87SiO_2 - 0.75C_3S$.
- $C_3A \approx 2.65Al_2O_3 - 1.69Fe_2O_3$.
- $C_4AF \approx 3.04Fe_2O_3$.

Properties of Cement

- Fineness: finer $\rightarrow$ faster hydration, early strength, more heat.
- Heat of hydration: exothermic reaction; important in mass concrete.
- Types: OPC, PPC, PSC, SRPC (sulfate resisting), RHPC (rapid hardening).

Aggregates

- MSA: limited by member size; affects workability & strength.
- Fineness modulus (FM): sum of % retained / 100, indicates coarseness.
- Moisture states: OD, AD, SSD (reference state), Wet.
- Moisture calc: $Absorption = (W_{ssd} - W_{dry}) / W_{dry} \times 100\%$.
- Surface moisture = Total – Absorption $\rightarrow$ adjust batch water.

Admixtures

- Plasticizers: 5–15% water reduction.
- Superplasticizers (PCE, SNF, SMF): 25–35% water reduction.
- Accelerators: e.g., $CaCl_2$ (avoid in RC).
- Retarders: e.g., sugar, lignosulfonates (delay set).
- Air entrainers: improve freeze-thaw resistance (3–6% air), reduce strength.

SCMs (Mineral Admixtures)

- Fly ash: Class F (low Ca, pozzolanic), Class C (higher Ca, cementitious).
- GGBS: reduces heat, improves durability, sulfate resistance.
- Silica fume: ultrafine, improves strength/durability, lowers workability.
- Rice Husk Ash: amorphous silica, variable quality.

Workability & Tests

- Workability = consistency + cohesiveness.
- Slump test: cone 300 mm, measure settlement in mm (IS 1199).
- Vebe test: time (s) for shape under vibration; stiff mixes.
- Compacting factor: $\text{wt}(\text{partially compacted})/\text{wt}(\text{fully compacted})$.
- Factors: water content, aggregate grading/shape, admixtures, cement fineness, time/temp.

Fresh Concrete Issues

- Segregation: separation of particles/paste.
- Bleeding: water rising to surface, weak surface layer.
- Control: proper grading, admixtures, lower w/c, careful placing/vibration.